

Use of National Surgical Quality Improvement Program Data as a Catalyst for Quality Improvement

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- BACKGROUND:** Semiannually, the National Surgical Quality Improvement Program (NSQIP) provides its participating sites with observed-to-expected (O/E) ratios for 30-day postoperative mortality and morbidity. At each reporting period, there is typically a small group of hospitals with statistically significantly high O/E ratios, meaning that their patients have experienced more adverse events than would be expected on the basis of the population characteristics. An important issue is to determine which actions a surgical service should take in the presence of a high O/E ratio.
- STUDY DESIGN:** This article reviews case studies of how some of the Department of Veterans Affairs and private-sector NSQIP participating sites used the clinically rich NSQIP database for local quality improvement efforts. Data on postoperative adverse events before and after these local quality improvement efforts are presented.
- RESULTS:** After local quality improvement efforts, wound complication rates were reduced at the Salt Lake City Veterans Affairs medical center by 47%, surgical site infections in patients undergoing intraabdominal surgery were reduced at the University of Virginia by 36%, and urinary tract infections in vascular patients were reduced at the Massachusetts General Hospital by 74%. At some sites participating in the NSQIP, notably the Massachusetts General Hospital and the University of Virginia, the NSQIP has served as the basis for surgical service-wide outcomes research and quality improvement programs.
- CONCLUSIONS:** The NSQIP not only provides participating sites with risk-adjusted surgical mortality and morbidity outcomes semiannually, but the clinically rich NSQIP database can also serve as a catalyst for local quality improvement programs to significantly reduce postoperative adverse event rates. (*J Am Coll Surg* 2007;204:1293-1300. © 2007 by the American College of Surgeons)

The National Surgical Quality Improvement Program (NSQIP) collects preoperative demographics and comorbidities, operative data, and 30-day postoperative mortality

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and morbidity outcomes on a systematic sample of patients undergoing major operations at participating medical centers. The preoperative data are used to develop prediction models for 30-day postoperative mortality and morbidity.¹⁻³ Observed-to-expected (O/E) mortality and morbidity ratios and 90% confidence intervals for mortality and 99% confidence intervals for morbidity are calculated for each medical center. These risk-adjusted outcomes are fed back to the participating medical centers every 6 months so that the medical centers can compare their performance with those in other centers in the program. Validation studies have shown that the O/E ratios are indicators of quality of surgical care.^{4,5}

If the lower limit of the confidence interval for the O/E ratio is greater than 1.0, this means that the center's patients are experiencing more adverse outcomes than would be expected on the basis of the patient characteristics. If the upper bound of the confidence interval is less than 1.0, this means that the center's patients are experiencing fewer adverse outcomes than would be ex-

Abbreviations and Acronyms

ACS	= American College of Surgeons
ASA	= American Society of Anesthesiologists
BMI	= body mass index
MGH	= Massachusetts General Hospital
NSQIP	= National Surgical Quality Improvement Program
O/E	= observed-to-expected
SSI	= surgical site infection
UTI	= urinary tract infection
UVA	= University of Virginia
VA	= Veterans Affairs

pected on the basis of the patient characteristics. If the confidence interval includes 1.0, this means that the center's patients are experiencing an adverse event rate that would be expected on the basis of the patient characteristics. Most medical centers have O/E ratios that are in the middle, that is, are not statistically significantly high or low. In each reporting period, there are a few medical centers that have statistically significant low O/E ratios (a desirable outcome) or statistically significant high O/E ratios (an undesirable outcome).

In the Veterans Affairs (VA) NSQIP, data have been collected for a long time on which processes and structures of care might have resulted in low mortality O/E ratios, and these data on "best surgical practices" have been shared with all participating VA medical centers in the VA NSQIP annual reports. In the case of medical centers with high-mortality O/E ratios, paper audits of deaths have been performed and reviewed by personnel in the VA Central Office and by the VA NSQIP Executive Board. If a medical center has had high-mortality O/E ratios in several consecutive reporting periods, site visits have been made by teams of surgeons, anesthesiologists, and surgical nurses.⁶⁻⁹

The most vexing situation is when a medical center has a high O/E ratio for 30-day postoperative mortality or morbidity. The resulting questions are: What should the surgical service do about the high O/E ratio? How should the situation be investigated to determine the causative factors underlying the high O/E ratio among the many surgical processes and structures that are involved in the care of the surgical patients and their risk-adjusted outcomes?

The purpose of this article is to describe some investigations and maneuvers accomplished by surgical services participating in the NSQIP that have used their clinically rich, local NSQIP database to try to explain

their high O/E ratios and to improve their risk-adjusted surgical outcomes. In the private-sector NSQIP, which has been made available to qualified sites, the American College of Surgeons (ACS) NSQIP data collection is accomplished through an Internet-based system that supports many online reports available to the participating centers to help them in this effort. These online reports will be described as well.

METHODS

Patients included in the NSQIP are those undergoing major operations under general, spinal, or epidural anesthesia. In the VA, eight major surgical subspecialties are represented (general, vascular, orthopaedics, urology, thoracic, otolaryngology-head and -neck surgery, neurosurgery, and plastic surgery); in the private sector, data was originally limited to general and vascular surgery but has been recently expanded to include the eight subspecialties captured by the VA and two additional subspecialties, neurosurgery and cardiac surgery in a multispecialty participation option. Exclusions include operations with known low postoperative mortality and morbidity, defined by specific current procedural terminology (CPT) codes. To ensure a representative sample of surgical patients from all days of the week, the first 40 (36 in the VA) consecutive, eligible surgical patients are assessed in each 8-day cycle, with each cycle starting on a different day of the week. Certain very high volume operations, such as inguinal hernia repairs and breast lumpectomies, are limited to the first 5 in each 8-day cycle so that these operations do not overwhelm the database.

Preoperative, operative, and postoperative variables in the data collection were selected on the basis of clinical relevance, reliability of data collection, and availability and ease of data collection. Preoperative variables include demographics; some lifestyle variables; functional status; American Society of Anesthesiologists (ASA) classification; selected laboratory tests; and selected pulmonary, cardiac, hepatobiliary, renal, vascular, central nervous system, nutritional, and immunologic comorbidities. Operative data include CPT codes of the primary operation and secondary operations, emergency status, wound classification, anesthesia method, operative times, and blood loss and replacement. Outcome variables include 30-day mortality from any cause inside or outside the hospital, length of stay, return to the operating room, and 19 different postoperative complications occurring in the 30-day postoperative period. An

overall morbidity score is computed by considering whether patients have had no postoperative complications versus whether they have had one or more complications. Because the intention of the NSQIP was to cover all major operations performed in a surgical service, the design of the program called for inclusion of mostly generic variables and few disease-specific or operation-specific variables.

A surgical clinical nurse reviewer is assigned at each medical center to collect the NSQIP data. The nurses receive in-depth training on the protocol, patient selection, definition of variables, and data collection methods. An operations manual that outlines these procedures in depth is also provided to each nurse. Regular conference calls and annual nurse meetings are held to maintain data uniformity and accuracy. The nurses are coordinated by national nurse clinical specialists. Periodically, interrater reliability site visits are conducted, in which a national nurse clinical specialist visits the site, reviews the case selection, reabstracts a sample of charts, and compares results with the locally obtained variable values.

The semiannual risk-adjusted outcomes reports include data on characteristics of participating hospitals; statistical prediction models for mortality and morbidity; the total number of assessed patients; observed number of postoperative adverse events; the observed adverse event rate; the number of expected adverse events; the expected adverse event rate; the O/E ratio; and the 90% (for mortality) or 99% (for morbidity) confidence intervals for the O/E ratios for each site, for mortality and overall morbidity, and for each surgical specialty, and all surgical specialties combined; the O/E ratios for each site over time; a comparison of patient characteristics between those who had an adverse event and those who did not; the highest frequency CPT codes in the database for each subspecialty; the number of patients excluded from the calculation of the O/E ratios and reason for exclusion by site; the interrater reliability site visit results; and patient followup rates. A glossary of terms, answers to frequently asked questions, and the NSQIP bibliography are also presented in the semiannual report. The risk-adjusted surgical outcomes are blinded by using site numbers, so that each site knows its own results but not those of other centers.

American College of Surgeons NSQIP participating sites in the private sector also have access to informative on-line reports on the ACS NSQIP website. These on-

line reports include displaying data related to preoperative risk factors, postoperative adverse events, mortality events, surgical site infections by preoperative wound classification, changes in patient variables over time, case volume at the site and types of procedures, and some physician-specific reports. Sites can benchmark their data with the national data by specifying surgical specialties or CPT codes, date ranges, and hospital types and bed sizes, and then producing reports comparing their patient preoperative characteristics and postoperative outcomes with the national averages. They can also "drill down" into their own site's database to obtain listings of patients who have experienced postoperative adverse events, or rates of adverse events for specific CPT codes.

RESULTS

Program-wide and site-specific quality initiatives in the presence of statistically significant high O/E ratios will be described in this section. Different initiatives will be described for high O/E ratios for mortality and high O/E ratios for morbidity.

High observed-to-expected mortality ratios

The first action to take in the presence of a high O/E mortality or morbidity ratio is to review the quality of the site's data. A comparison of the site's patient characteristics with the national average can reveal potential underreporting of comorbidities at the site or systematic biases in determining such important predictor variables as ASA class, emergency status, or functional status. If important differences are found between the site's patient characteristics and the national averages, there should be a thorough review of data collection methods and data definitions. Underreporting of comorbidities could lead to an underestimate of the expected adverse event rate and an overestimate of the O/E ratio. But it must be realized that these differences could also reflect true differences in the site's patient population compared with the overall NSQIP population at all participating sites, and these differences are appropriately adjusted for in the risk-adjusted outcomes.

The next action to take should be a review of all postoperative deaths in the NSQIP database at the site. For each of these patients, the probability of death from the prediction model using the patient preoperative characteristics should be computed, and emphasis should be placed on reviewing surgical processes and structures of care, particularly for patients who had a low probability

of death but who died within 30 days after operation. Review of these patients by several independent surgeon reviewers (outside of the site's surgical service) could be useful in obtaining an unbiased opinion of the quality of care given to the patients.

In the VA NSQIP, there was a recent review of the episode of care for all deaths in patients having colorectal operations between 2003 and 2005 in VA medical centers, who had high O/E mortality ratios for colorectal surgery, or who had 30-day unadjusted mortality rates that were more than twice the VA national average of 5.9%.¹⁰ For each reported colectomy death, the medical centers submitted a copy of the operative report, death summary, mortality and morbidity evaluation report, and completed a questionnaire pertaining to the perioperative care for each patient, from diagnosis to death. The data were divided among four VA NSQIP general surgeon board members for review. The results were then tabulated based on patient characteristics, disease, provider or system-based issues, and choices of operation. A total of 103 patients from 8 medical centers were reviewed. The mean patient age was 74 years, with 63% of the patients undergoing emergency operations. Fifty-four percent of the patients had malignant disease and 21% had metastatic disease. The reviewers judged that 19% of the patients had a delay in diagnosis, that 22% of the patients should have been operated on earlier, and that 14% of the patients should have had a different operation, usually less radical. In addition, the reviewers identified system-related issues in 19% of the patients and practitioner-related issues in 20% of the patients. Based on these findings, the high-outlier hospitals were able to address the specific issues pertinent to their system within the continuum of care for colorectal surgery patients. It is obvious from this example that system and structure issues related to timely referral, proper diagnosis, and referral to surgery are as important as the choice of operation and postoperative care.¹⁰

High observed-to-expected morbidity ratios

Because the main morbidity outcome variable in the NSQIP is whether or not the patient has experienced 1 or more 30-day postoperative complications, in the presence of a high O/E morbidity ratio, the first action item of the site should be to compare the site's unadjusted morbidity rate with the national average for each of the individual 19 NSQIP postoperative complications. If a site is particularly high for certain specific complica-

tions, this is a good place to start quality improvement activities.

The Salt Lake City VA experience

In 1996, the Salt Lake City VA medical center found that it had a high morbidity O/E ratio (1.39) in general surgery. Initial data analysis revealed that the patients had high rates of wound complications, postoperative ileus or bowel obstruction, and need for ventilatory support for more than 48 hours after surgery. Chart review of patients with postoperative ileus or bowel obstruction and need for ventilatory support revealed no issues with providers or standards of care. Chart review of patients with wound complications revealed a practice pattern of closing contaminated or clean-contaminated wounds in the presence of other risk factors for wound complications. A wound infection prevention protocol (Table 1) was put into place based on this analysis and a review of the literature. The wound complication rate dropped from 5.5% in 1996 to 2.9% in 1998 ($p = 0.018$), and the morbidity O/E ratio for general surgery at the Salt Lake City VA medical center was reduced from 1.39 in 1996 to 1.19 in 1998.¹¹

The University of Virginia experience

Program participants at the University of Virginia (UVA), Charlottesville, used their ACS NSQIP data as part of a Virginia Health Care Quality Center, Surgical Infection Prevention (SIP) initiative to reduce the incidence of surgical site infection (SSI) rates for intraabdominal operations. Using the ACS NSQIP website Wound Classification Occurrence Detail Report, UVA identified that from June 1, 2003, through April 30, 2004, 93% (67 of 72) of patients with clean-contaminated wounds who developed an SSI had undergone an intraabdominal procedure. Using the ACS NSQIP online CPT Volume Analysis Report, the most frequently performed colorectal procedures at UVA were identified (CPT codes 44140, 44625, 44145, 44160, 44143, 44146, 44320, 44626, 44130, and 44150). The rate of SSI in these procedures at UVA from June 1, 2003 to April 30, 2004, was 17.6%, compared with the national average of 8.1% ($p = 0.0002$).

Detailed information from the ACS NSQIP Web site on-line Select Postoperative Occurrence Report allowed for further analysis of the specific characteristics of patients having SSI versus those who did not have SSI. This comparison revealed that the subset of colorectal pa-

Table 1. Wound Infection and Disruption Prevention Protocols at the Salt Lake City Department of Veterans Affairs Medical Center

Wound infection	Wound disruption
If the patient has three or more of the following risk factors, the skin should be left open to heal by secondary intention or delayed primary closure:	If the patient has three or more of the following risk factors, consider measures to prevent wound disruption, such as retention sutures (internal or external) or placement of Dexon (Syneture) mesh:
1. Emergency status	1. Emergency status
2. Immunosuppression	2. Immunosuppression
A. Cancer	A. Cancer
B. Transplant	B. Transplant
C. On steroids	C. On steroids
3. Open gastrointestinal tract	3. Open gastrointestinal tract
4. Obesity	4. Obesity
5. Diabetes	5. Diabetes
6. Alcohol/tobacco/drug abuse	6. Chronic obstructive pulmonary disease
7. Pus/stool/anything but clear fluid in abdomen	7. Ileus/bowel obstruction preoperatively
8. Homelessness	8. Preoperative albumin < 3.0 g/dL
9. Sepsis	9. Pus/stool/anything but clear fluid in abdomen

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tients who experienced an SSI tended to have, on average, a higher body mass index (BMI) than patients with no SSI (29.1 versus 27.0 kg/m², $p = 0.0981$). This trend was also observed for the ACS NSQIP patients as a whole (28.4 versus 27.2 kg/m², $p = 0.077$). The UVA NSQIP Colorectal Subcommittee corroborated that in its colorectal patient population, a BMI > 25 kg/m² placed a patient at risk for developing an SSI after surgery.¹² The ACS NSQIP reports enabled UVA to target its interventions by being informed with very specific data about patients in whom the SSI was occurring and to develop appropriate changes to the clinical care of those patients.

As a result of this analysis, wound wicking, in which a piece of dry gauze is placed in the wound to wick exudates away from the area, in colorectal patients with a BMI > 25 kg/m², was implemented at UVA. Additionally, a new protocol which included Surgical Infection Prevention measures to prevent SSI in abdominal surgery (CPT codes 43000 to 48999), focused on administration of proper prophylactic antibiotics 0 to 60 minutes before incision, discontinuation of antibiotics after 24 hours, maintenance of intraoperative normothermia; optimal glycemic control in diabetic patients was also put into place (Table 2). Overall SSI infection rates for a subset of colorectal patients with the CPT codes mentioned before decreased from 17.6% from June 1, 2003, to April 30, 2004, to 12.7% in the period from June 1, 2004, to April 30, 2005 ($p = 0.128$), and to 11.2% in the period from June 1, 2005, to April 30, 2006 ($p = 0.051$). This represents a decrease in the rate of SSI in patients undergoing intraabdominal procedures at

UVA for the procedures of interest of 36% and compares with a 15% increase in the rate of SSI in these patients in the ACS NSQIP as a whole for the same time periods.

Analysis by UVA of the online Web site reports and data from the NSQIP allowed identification of a problem with SSI in colorectal patients with a BMI > 25 kg/m². Further review resulted in development of new clinical protocol, which included wound wicking, and adoption of the Surgical Infection Prevention guidelines, which resulted in clinically significant reductions in SSIs. The online reports enable UVA to continue to monitor SSI rates over time and, if necessary, to review and update changes to clinical care in a timely manner.

The Massachusetts General Hospital experience

The Massachusetts General Hospital (MGH)—Codman Center for Clinical Effectiveness in Surgery (CCCES) was established to conduct surgical outcomes research that will improve the quality, patient safety, and results of surgical care. Using the ACS NSQIP data as the foundation for their work, the Center has identified opportunities to improve its surgical outcomes and resident education mortality and morbidity conferences.

Table 2. Surgical Site Infection Wound Infection Prevention Protocol for Colorectal Patients with Body Mass Index > 25 kg/m² at the University of Virginia Health System

1. Wound wicking
2. Administration of proper prophylactic antibiotics 0–60 minutes before incision
3. Discontinuation of antibiotics after 24 hours
4. Maintenance of intraoperative normothermia
5. Optimal glycemic control for diabetic patients

The MGH vascular surgery O/E morbidity ratio for the surgery dates from October 1, 2002, to September 30, 2003, revealed that the number of observed morbidities was slightly higher than would have been expected given the patient preoperative risk factors and types of operations performed (1.19, 99% CI 0.93 to 1.48). Although the O/E ratio was not statistically significantly higher than 1.0 according to NSQIP standards, MGH decided to explore ways to reduce it. Using the program's online reports, MGH identified that one of the largest factors for the high O/E ratio was a high rate of urinary tract infections (UTIs) compared with the other sites participating in the NSQIP (7.0% versus 4.7%, $p = 0.087$). Working with the multidisciplinary Clinical Performance Management Team for Surgery and using documents and information about UTIs that have subsequently been posted and made available on the program website, the group identified general and detailed clinical information about UTI that enabled them to understand the scope of the problem, including estimates of the associated costs of caring for patients who develop a UTI after surgery. UTIs account for up to 40% of nosocomial infections,^{13,14} with urinary catheter-related infections causing the vast majority of nosocomial UTIs.¹⁵ Each hospital-acquired UTI adds approximately \$675 to the costs of hospitalization.¹⁶ When bacteremia develops, this additional cost increases to at least \$2,800.¹⁶ This information and cost data were important to the group as they sought to secure the appropriate hospital resources required to implement changes to clinical care.

A metaanalysis of studies of silver alloy and antibiotic-impregnated urethral catheters that included four randomized clinical trials comparing the efficacy of silver catheter versus standard, noncoated catheters was also completed using information posted on the ACS NSQIP program Web site.

The four clinical trials all showed significant reduction in the development of catheter-associated bacteriuria.¹⁷⁻²⁰ Data supporting the use of silver alloy urinary catheters to reduce urinary catheter-related bacteriuria were found to be reasonably strong. Based on that information and other clinical studies posted on the program Web site, changes to physician order entry templates, the Foley catheter removal algorithm, the use of silver-coated catheters for select patients, identification of procedures not requiring a catheter, and an educational campaign for clinicians were implemented.

The result of this initiative was a significant reduction in the vascular UTI rate at MGH from 7.0% in the period October 1, 2002, to September 30, 2003, to 1.8% from October 1, 2003, to September 30, 2004 ($p = 0.0004$), and a reduction in the vascular O/E morbidity ratio from 1.19 (99% CI 0.93 to 1.48) to 0.93 (99% CI 0.67 to 1.48). MGH continues to monitor the effectiveness of these interventions using the risk-adjusted morbidity reports and online reports and to watch for any unfavorable trends in its UTI rate.

In another Center initiative, a retrospective study was performed comparing data from the morbidity and mortality conference in general surgery, in which complications and deaths were identified by residents or attending surgeons, with ACS NSQIP data compiled by the MGH nurse reviewer from July 1, 2002 to June 30, 2003.²¹ Mortality rates calculated by the traditional morbidity and mortality conference compared with the MGH ACS NSQIP data were 0.9% versus 1.9%. Complication rates reported in morbidity and mortality conferences were 6.4% versus 28.9% in the NSQIP. Approximately one of two deaths and three of four complications were not reported in the MGH morbidity and mortality conference. As a result of this study's findings, a Web-based reporting system based on the ACS NSQIP platform was created to automate, facilitate, and standardize data on surgical morbidity and mortality and to improve resident surgical education at MGH. A major goal of this new reporting system is to educate and familiarize both residents and attending physicians with quality assessment, and the importance of reliable data made possible by standardized definitions, sound statistics, and responsible conclusions.

DISCUSSION

The purpose of this article was to illustrate ways in which the rich NSQIP database has been used locally at participating sites in both the VA and the private sector for surgical quality improvement activities. Many of these activities were prompted by a site's observation that it had a high mortality or morbidity O/E ratio, although a site does not have to wait for that situation to put the data to good use, as is shown in the examples from Massachusetts General Hospital. Interventions implemented at the participating sites in many cases resulted in statistically significant reductions in postoperative morbidity.

At some sites, the NSQIP has served as the basis for a surgical service-wide outcomes research and quality im-

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provement center. This is illustrated best by the activities at Massachusetts General Hospital and the University of Virginia at Charlottesville. These centers are important because they promote and encourage interest among surgeons in the use of clinical and outcomes research to improve surgical care.

In the Department of Veterans Affairs, there has been a reduction in overall 30-day postoperative mortality from 3.16% in Phase 1 (October 1, 1991 to December 31, 1993) to 1.70% in fiscal year 2005 ($p < 0.0001$). There has also been a reduction in overall 30-day postoperative morbidity in this same time interval, from 17.44% in Phase 1 to 10.18% in fiscal year 2005 ($p < 0.0001$). In the Patient Safety in Surgery Study, which ran from October 1, 2001, to September 30, 2004, we did not observe a significant reduction in 30-day postoperative mortality, but we did see a statistically significant reduction in 30-day postoperative morbidity (-8.7%, $p = 0.002$). Although we have not performed a randomized clinical trial to prove that these observed reductions are from the NSQIP intervention, local quality improvement efforts, as illustrated in this article, have probably contributed to the observed reductions.

The philosophy of the NSQIP is not to point out "bad apples" among surgical programs, but rather to provide surgeons with highly reliable data that they can believe in and that will empower them to make quality improvement efforts. Some investigators have devoted considerable energy to developing reporting systems based on administrative data or a combination of administrative data and selected clinical data to enable surgical programs and administrators to compare risk-adjusted surgical outcomes.²² One reason they have made these efforts is because they believe that a system like the NSQIP, which requires trained nurses at each site to collect top-quality clinical data, is too costly. But some studies have shown the poor quality of administrative data compared with good-quality clinical data collected by trained personnel at the participating sites.²³ In addition, complications are costly and can raise the median cost of hospitalization for major surgery by up to fivefold.²⁴ The savings realized by reducing complication rates through local quality improvement initiatives, such as those described in this article, far exceed the cost of a salary of a single nurse. Those who advocate the use of administrative data for risk adjustment in lieu of the NSQIP have, to some extent, missed the point of the value of a system like the NSQIP. The NSQIP not only

provides a highly reliable data system to compare risk-adjusted outcomes that surgeons can have confidence in, but, as illustrated in this article, it also provides robust data to allow for intensive quality improvement efforts at the local level.

Author Contributions

Study conception and design: Khuri, Henderson, Rowell

Acquisition of data: Rowell, Turrentine, Hutter

Analysis and interpretation of data: Rowell, Turrentine, Hutter, Henderson, Khuri

Drafting of manuscript: Rowell

Critical revision: Henderson, Rowell, Turrentine, Hutter

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